Reflection in a Plane Mirror

Aim: To investigate the law of reflection

Equipment:

Hodson light box	Plane mirror
Single Slit Slide	Ruler
Power pack	Protractor
Sharp pencil	

Diagram of Equipment Set-up

Procedure

1. Place the single slit slide in the end of the Hodson light box closest to the light source.
2. Connect the Hodson light box to the AC terminals of the power pack, set the voltage to 12 V and plug the power pack into the wall.
3. Place the mirror on the line below.
4. Shine the light ray onto the mirror so that it hits the mirror at the black dot.
5. Mark the path of the incident ray and reflected ray by placing three dots along each ray, then use a ruler to connect the dots.
6. Mark the incident ray, i_1, and the reflected ray, r_1.
7. Repeat this for 5 other angles, making sure the light ray hits the mirror at the cross and labelling the incident and reflected rays as you go. (i_2, r_2, i_3, etc)

Results

Processing of Results

1. Draw a line perpendicular to the mirror, starting at the black dot. This is the normal.
2. Measure the angle between the normal and incident ray 1 and record it in the table.
3. Measure the angle between the normal and reflected ray 1 and record it in the table.
4. Repeat for all rays of incidence and reflection.

	Angle of Incidence	Angle of Reflection
Ray 1		
Ray 2		
Ray 3		
Ray 4		
Ray 5		
Ray 6		

Questions

1. What is the Law of Reflection?

2. Do your experimental results support the Law of Reflection? Give a reason for your decision.

3. Describe ways in which you could improve the results of your experiment.

4. Give two examples where you have observed the Law of Reflection in action.